

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA**

M. Sc. (BC/ BT/ MI) Sem - I Examination (Backlog), 2011-12

BC/ BT/ MI - 702 Cell biology

Date: 17. 04. 12, Tuesday

Time: 10.00 AM to 1.00 PM

Maximum Marks: 70

Instructions:

1. Provide all necessary information as required in the answer sheet.
2. Attempt all questions.
3. Questions in **Part A** should be answered in the question paper itself. Please **do not write** your candidate ID number on the question paper for Part A.
4. Write answers for Section I and Section II of **Part B** in separate answer sheets.
5. Use of cell phones is strictly prohibited.
6. Use of non-programmable calculators may be allowed if needed.

PART A

Total Marks: 20

Q.1 Choose the correct option

1. In desmosomes, cadherins link to ----- of an adjacent cell
 - (a) Integrins
 - (b) Intermediate filaments
 - (c) Plasmodesmata
 - (d) Rab Proteins
2. Which of the following is **NOT** true for K^+ channels?
 - (a) They are ion gated channels
 - (b) They function as a homotetramer
 - (c) A conserved amino acid sequence forms the ion selectivity filter
 - (d) None of the above
3. Glucose uptake in the intestinal lumen *via* $2Na^+$ -glucose symporter is an example of
 - (a) Active transport
 - (b) Channel mediated transport
 - (c) ATP-driven transport
 - (d) Secondary active transport
4. Fibronectin in extracellular matrix can be best characterized as
 - (a) Proteoglycan
 - (b) Structural fibrous protein
 - (c) Adhesive fibrous protein
 - (d) None of the above
5. Transmembrane linker proteins in hemidesmosomes belong to ----- family of extracellular matrix receptors
 - (a) Cadherins
 - (b) Integrins
 - (c) Laminins
 - (d) None of the above
6. ----- is the disaccharide composed of D-glucuronic acid and N-acetyl- D- Galactosamine
 - (a) Chondroitin sulfate
 - (b) Heparan sulphate
 - (c) Keratan sulphate
 - (d) None of the above
7. Which of the following is not a direct function of Na^+-K^+ ATPase?
 - (a) Maintaining membrane potential
 - (b) Osmosis regulation
 - (c) Transport of Na and K ions
 - (d) None of the above
8. All the organelles are double membranous structure except
 - (a) Mitochondria
 - (b) Chloroplast
 - (c) Endoplasmic reticulum
 - (d) Lysosome

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9. Find out the incorrect pairing of translocator protein and organelle

(a) Calnexin	—	Endoplasmic reticulum
(b) Tim22	—	Mitochondria
(c) Neucleoporphin	—	Nucleus
(d) Par (peroxin)	—	Peroxisomes
10. Mitotic spindle fiber is

(a) Myosin fiber	(b) Microtubule
(c) Microfilament	(d) None of above
11. Name the detoxifying organelle in the liver which proliferate to decrease the harmful effects of alcohol

(a) Lysosome	(b) Peroxisome
(c) Rough endoplasmic reticulum	(d) Smooth endoplasmic reticulum
12. ----- link one signaling protein to another, but do not convey the signal themselves

(a) Adaptor proteins	(b) Scaffold proteins
(c) Relay proteins	(d) Second messengers
13. ----- is a minor phospholipid in the inner leaflet of plasma membrane bilayer, produced by phosphorylation of phosphatidyl-inositol and involved in cell signaling

(a) PIP2	(b) Phospholipase C
(c) Calmodulin	(d) Diacylglycerol
14. Wnt proteins bind to Frizzled receptors and inhibit the degradation of -----

(a) Hedgehog	(b) Notch
(c) β catenin	(d) All the above
15. α subunit of a G protein

(a) Binds to receptors but do not activate them	(b) Binds GDP in inactive form and GTP in active form.
(c) Is receptor for insulin	(d) All the above
16. ----- begins when pairs of sister chromatids align in the center of the cell

(a) Anaphase	(b) Interphase
(c) Metaphase	(d) Telophase
17. The G0 phase is

(a) The checkpoint before G1	(b) The state of most cells in an animal body
(c) Another name for interphase	(d) None of the above
18. ----- protein functions both as an E3 ubiquitin ligase and an inhibitor of p53 transcriptional activation.

(a) ERK	(b) Mdm2
(c) Bax	(d) Chk1
19. Which of the following is/are the proteolytic target/s of effector caspases

(a) Actin	(b) PARP
(c) Nuclear Lamins	(d) All the above
20. Photorespiration is the process of incorporation of molecular oxygen instead of carbon dioxide during the carbon fixation pathway

	[True] [False]
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PART B

Total Marks: 50

SECTION I

- Q.1** (A) Explain using a neatly labeled diagram, the different ways in which proteins can be associated with the phospholipid membrane **04**
- OR**
- (A) Explain the molecular transport mechanism underlying treatment of diarrhea using ORS (mixture of salt and sugar). **04**
- (B) **Explain the following (Any two)** **06**
- (i) Different functions of ion channels
 - (ii) Role of cholesterol in the cell membrane
 - (iii) Patch clamp technique
- Q.2** (A) **Answer the following (Any two)** **06**
- (i) What is compartmentalization? Elucidate the salient features of each compartment. Why is it essential for normal cellular function? explain with appropriate example
 - (ii) Discuss mis folded protein response
 - (iii) Enlist the name of organelles which are engaged in protein synthesis and explain any one in detail
- (B) **Answer the following (Any two)** **04**
- (i) Structure and functions of peroxisomes
 - (ii) Role of Golgi body in protein sorting
 - (iii) Nucleopore complex

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SECTION II

- Q.3** (A) Discuss different types of ion-linked receptors 04
- OR**
- (A) Briefly describe how a G protein coupled receptor works 04
- (B) **Briefly discuss the following (Any two)** 04
- (i) cAMP production and degradation
- (ii) Ras protooncogenes
- (iii) Jun-N-terminal kinase
- (C) Explain the JAK-STAT signal transduction mechanism for the erythropoietin receptor 02
-
- Q.4** (A) Discuss the different types of cell-cell junctions 04
- OR**
- (A) Explain the role of cell adhesion molecules in cellular communication 04
- (B) **Answer the following (Any two)** 06
- (i) What are proteoglycans? Explain the role of hyaluronan in extra cellular matrix.
- (ii) Ca^{2+} independent Cell-Cell Adhesion
- (iii) Briefly describe the structure and functions of Plasmodesmata
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- Q.5** (A) Explain the differences between apoptosis and necrosis 04
- OR**
- (A) Briefly discuss the major cyclins and Cdk's of vertebrates and its functions 04
- (B) **Answer the following (Any two)** 04
- (i) Two hit mutation model for retinoblastoma
- (ii) Origin Recognition Complex (ORC)
- (iii) Cell cycle checkpoints
- (C) Discuss the extrinsic pathway of apoptosis 02